

BRIDGE-relay

Software for Universal Wireless Interface for Bionic Limbs Trainee Project

Purpose

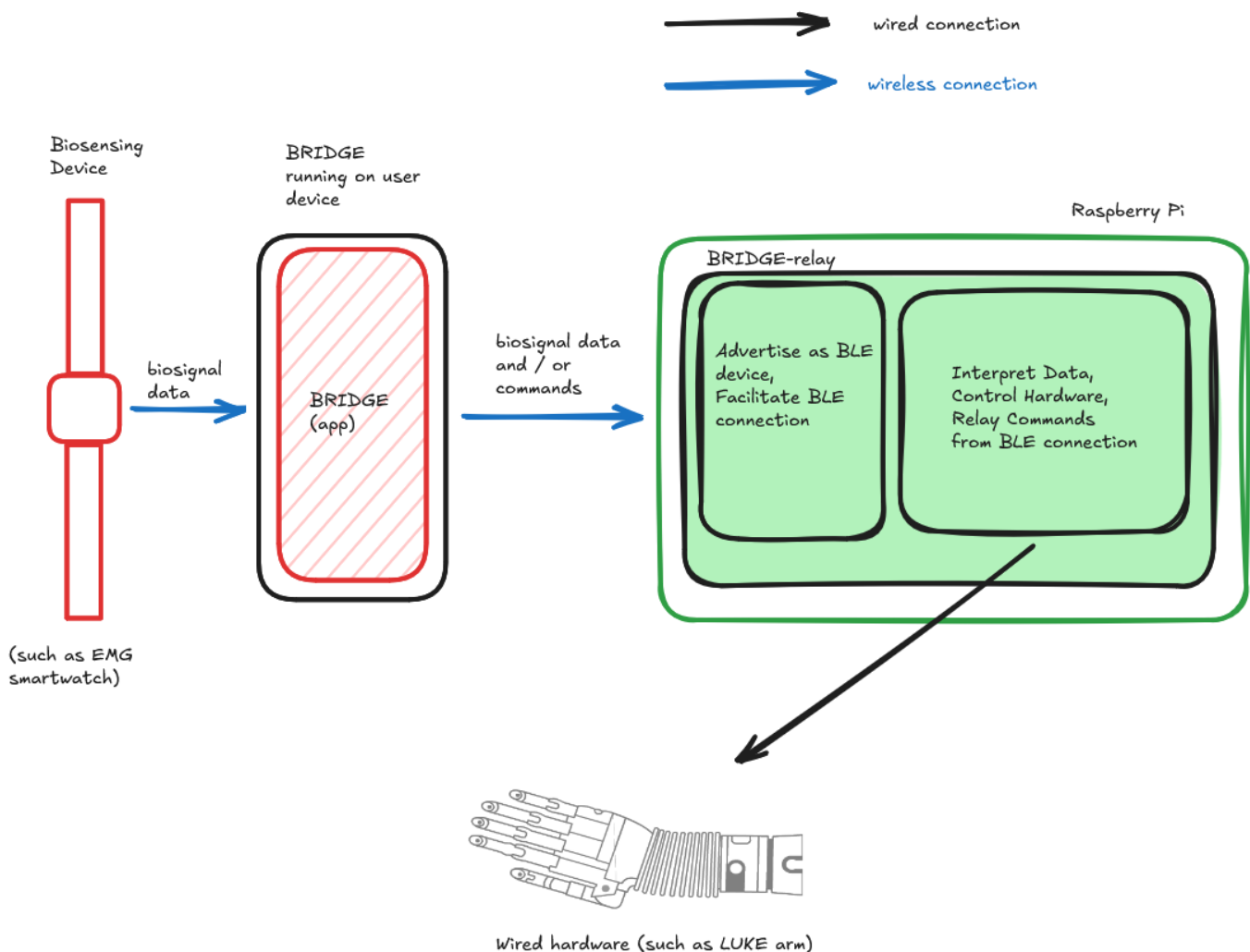
The BRIDGE-relay is designed to be an interface between [BRIDGE](#) and lab hardware typically only controlled via wired connections. Such as the DEKA "Luke" Hand.

Architecture

BRIDGE-relay is simple in its composition.

BRIDGE-relay is, in short, two programs communicating with eachother using local/internal TCP/IP connections to achieve the following:

- Advertise a device we want to control as a bluetooth device that BRIDGE can connect to
- Relay commands or data from BRIDGE (as collected in real-time from a biosensing device such as the EMG smartwatch) to the connected wired hardware (such as the Luke Hand).



Technology Stack

- Hardware
 - Raspberry Pi
 - (Optional) Bluetooth adapter
- Software
 - Linux (likely raspbian or something lighter...)
 - Python or C++ (preferably Python for ease-of-use)
 - Pyserial
 - BlueZ

Development

Template code has been added to `src/`. This is not the final software for this project and is subject to change.

Usage

1. Connect hardware device to BRIDGE-relay
2. Turn on BRIDGE-relay
3. On the BRIDGE app, in "Connections", select the option "connect to relay" and scan for bluetooth devices.
4. Connect to the device titled "BRIDGE_relay"
5. Connect your biosensing device and begin streaming data.

Example with EMG-smartwatch and LUKE Hand

1. Connect PCAN-USB connector from LUKE Hand to Raspberry Pi (BRIDGE-relay)
2. Turn on BRIDGE-relay
3. On the BRIDGE app, in "Connections", select the option "connect to relay" and scan for bluetooth devices.
4. Connect to the device titled "BRIDGE_relay"
5. A successful connection will return a success message along with the control software running on BRIDGE-Relay. The control software should read "DEMO-LUKE-HAND-CTRL"
6. Connect EMG smartwatch to begin streaming.
7. Grasping with the EMG smartwatch should cause the LUKE Hand to perform a grasp in real time.

Troubleshooting

- to be added once prototype is assembled